

GUERNSEY, Guernsey

WMO: 038940

Lat: 49.433N

Lon: 2.598W

Elev: 336

StdP: 14.52

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	33.5	35.5	23.8	18.2	36.3	26.3	20.5	37.7	37.7	46.5	33.2	46.1	14.1	90	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	74.4	64.2	71.1	62.9	68.5	61.9	65.7	71.3	64.3	68.7	63.2	66.8	9.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	89.3	67.3	62.6	86.0	65.7	61.6	83.1	64.7	30.6	71.6	29.6	68.9	28.8	66.8	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.9	27.5	24.8	30.8	81.3	3.0	4.9	28.6	84.8	26.9	87.7	25.2	90.5	23.0	94.1
			28.2	68.9	2.7	2.1	26.3	70.4	24.8	71.6	23.3	72.8	21.4	74.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	45.0	44.5	46.4	49.2	53.8	58.4	62.0	62.5	60.1	56.1	50.6	46.7
	DBStd	7.54	4.53	4.22	3.83	3.99	4.05	3.72	3.43	3.13	3.09	3.69	3.95	4.62
	HDD50	662	159	155	121	58	8	0	0	0	0	4	39	118
	HDD65	4415	620	573	578	474	346	202	106	91	150	276	432	567
	CDD50	1759	4	2	8	34	128	253	372	388	304	193	57	16
	CDD65	37	0	0	0	0	1	5	13	14	4	0	0	0
	CDH74	121	0	0	0	0	2	23	46	47	3	0	0	0
	CDH80	15	0	0	0	0	0	2	6	7	0	0	0	0
Wind	WSAvg	12.8	15.4	14.3	13.3	11.9	11.8	11.2	11.0	10.6	11.6	13.1	14.3	15.3
Precipitation	PrecAvg	33.0	3.7	3.0	2.2	2.1	2.0	1.8	1.6	2.2	2.1	3.6	4.3	4.4
	PrecMax	40.6	8.1	5.9	6.7	5.0	4.7	4.4	3.3	4.7	5.5	6.7	9.0	9.6
	PrecMin	24.6	0.5	0.6	0.2	0.0	0.3	0.2	0.3	0.4	0.2	0.7	1.2	1.0
	PrecStd	4.6	1.9	1.5	1.3	1.3	1.2	1.0	0.8	1.2	1.2	1.5	1.9	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	54.3	58.7	66.1	71.9	77.1	79.2	79.4	73.6	67.0	60.2	56.3
		MCWB	52.7	50.4	52.2	56.2	60.8	65.3	65.8	67.0	63.6	61.6	57.9	54.5
	2%	DB	53.0	52.2	54.8	60.7	66.7	71.4	73.8	73.6	69.4	63.8	58.3	55.3
		MCWB	51.6	50.0	50.3	53.6	58.7	62.3	63.8	64.9	63.0	60.2	56.4	53.9
	5%	DB	51.9	51.0	52.6	57.0	63.0	67.6	70.6	69.9	66.6	62.2	57.2	54.1
		MCWB	50.5	49.3	49.5	51.8	56.5	60.6	62.6	63.1	61.5	59.4	55.2	52.6
	10%	DB	50.9	50.0	51.2	54.6	60.0	64.6	68.0	67.4	64.6	61.0	56.1	53.0
		MCWB	49.4	48.5	48.8	50.6	55.2	58.9	61.8	62.0	60.3	58.5	53.9	51.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.5	52.1	52.9	58.0	62.8	66.8	67.1	68.3	65.7	62.9	58.9	55.6
		MCDB	53.7	52.8	56.7	64.1	69.0	75.3	75.0	76.9	70.5	65.1	59.3	55.8
	2%	WB	52.2	50.9	51.4	54.4	59.5	63.5	65.2	65.9	63.9	61.3	57.1	54.3
		MCDB	52.6	51.6	53.3	58.8	64.5	69.7	71.2	70.9	67.2	63.0	57.8	54.8
	5%	WB	51.0	49.8	50.4	52.7	57.5	61.3	63.8	64.4	62.7	60.2	55.7	53.1
		MCDB	51.6	50.6	51.9	55.9	61.8	66.0	68.6	68.1	65.3	61.7	56.8	53.9
	10%	WB	49.8	48.7	49.6	51.3	55.8	59.7	62.6	63.2	61.6	59.1	54.3	51.9
		MCDB	50.7	49.7	50.9	53.8	59.1	63.4	66.6	66.1	63.8	60.6	55.8	52.8
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.1	7.1	8.6	9.2	9.8	9.7	9.0	8.1	6.6	5.7	5.7
		MCDBR	5.6	6.5	8.6	12.2	13.9	14.9	14.4	13.2	10.4	7.2	5.9	5.6
		MCWBR	6.5	6.1	6.3	7.7	8.2	8.6	7.6	7.0	6.0	5.4	6.2	6.7
	5% WB	MCDBR	5.6	6.0	7.5	10.5	12.4	13.0	12.4	10.9	9.0	6.6	5.7	5.7
		MCWBR	6.9	6.7	6.3	7.3	7.9	8.2	7.1	6.6	6.2	5.9	6.5	7.0
Clear-Sky Solar Irradiance	taub	0.334	0.349	0.380	0.391	0.396	0.389	0.385	0.378	0.368	0.360	0.335	0.327	
	taud	2.411	2.406	2.333	2.330	2.371	2.418	2.430	2.472	2.477	2.474	2.450	2.376	
	Ebn at Noon	225	251	261	271	274	276	276	273	264	245	223	208	
	Edh at Noon	21	26	33	37	37	36	35	32	29	24	20	19	
All-Sky Solar Radiation	RadAvg	276	519	911	1475	1779	1936	1872	1564	1192	672	347	226	
	RadStd	21	64	96	121	128	190	100	120	83	46	31	26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page